Marmara University
Electrical and Electronics Engineering
Probability and Statistics
Problem Sets with Solutions #4

[1] Consider a random experiment for which $S=\{s_i: s_i=i,\ i=1,2,\dots,10\}$ and the outcomes are equally likely. If a random variable is defined as $X(s_i)=s_i^2$, find S_X and PMF. Repeat for $S=\{s_i: s_i=-3,-2,-1,0,1,2,3\}$.

Solution: Let $S=\{s_i: s_i=i,\ i=1,2,\dots,10\}$

s_i	1	2	3	4	5	6	7	8	9	10
$X(s_i)$	1	4	9	16	25	36	49	64	81	100

Hence, $S_X=\{1,4,9,16,25,36,49,64,81,100\}$.

$P_X(1)=P_X(4)=\dots=P_X(100)=\frac{1}{10}$ since s_i's are equally likely

Let $S=\{s_i: s_i=-3,-2,-1,0,1,2,3\}$

s_i	-3	-2	-1	0	1	2	3
$X(s_i)$	9	4	1	0	1	4	9

Hence, $S_X=\{0,1,4,9\}$

$P_X(0)=P[s_i=0]=\frac{1}{7}$

$P_X(1)=P[s_i=-1]+P[s_i=1]=\frac{1}{7}+\frac{1}{7}=\frac{2}{7}$

$P_X(4)=P[s_i=-2]+P[s_i=2]=\frac{1}{7}+\frac{1}{7}=\frac{2}{7}$

$P_X(9)=P[s_i=-3]+P[s_i=3]=\frac{1}{7}+\frac{1}{7}=\frac{2}{7}$ ☒

[2] A man is late for his job by $s_i=i$ minutes, where $i=1,2,\dots$. If $P[s_i]=(\frac{1}{2})^i$ and he is fined \$0.5 per minute, find the PMF of his fine. Calculate the probability that he will be fined more than \$10.

SOLUTION: Let the random variable X be defined as $X(s_i)=0.5s_i=0.5i$. Note that X represent the fine

$P_X(x_i)=P[X(s_i)=x_i]=P[0.5s_i=x_i]=$

$=P[s_i=2x_i]=(\frac{1}{2})^{2x_i},\quad x_i=0.5,1,1.5,2,\dots$

$$P[X>10]=\sum_{\{x_i:\ x_i>10\}}P(X=x_i)=\sum_{\{x_i:\ x_i\geq 10\}}\left(\frac{1}{2}\right)^{2x_i}=\left(\frac{1}{2}\right)^{21}+\left(\frac{1}{2}\right)^{22}+\left(\frac{1}{2}\right)^{23}+\left(\frac{1}{2}\right)^{24}+\dots$$

$$=\sum_{i=1}^{21}\left(\frac{1}{2}\right)^i=\frac{(\frac{1}{2})^{21}}{1-\frac{1}{2}}=\left(\frac{1}{2}\right)^{20}$$ ☒

[3] The maximum value of the binomial PMF occurs for the unique value $k = \lfloor (M+1)p \rfloor$, where $\lfloor x \rfloor$ denotes the largest integer less than or equal to x, if $(M+1)p$ is not an integer. If, however, $(M+1)p$ is an integer, then the PMF will have the same maximum value at $k = (M+1)p$ and $k = (M+1)p - 1$. For the latter case when $(M+1)p$ is an integer, you are asked to prove this result. To do so, first show that

$$P_X[k] / P_X[k-1] = 1 + \frac{(M+1)p - k}{k(1-p)}$$

Solution:

$$P_X[k] = \binom{M}{k} p^k (1-p)^{M-k}$$

Let $Q[k] = \dfrac{P_X[k]}{P_X[k-1]}$

$$Q[k] = \frac{\binom{M}{k} p^k (1-p)^{M-k}}{\binom{M}{k-1} p^{k-1} (1-p)^{M-k+1}} = \frac{\binom{M}{k}}{\binom{M}{k-1}} \frac{p^k}{p^{k-1}} \frac{(1-p)^{M-k}}{(1-p)^{M-k+1}}$$

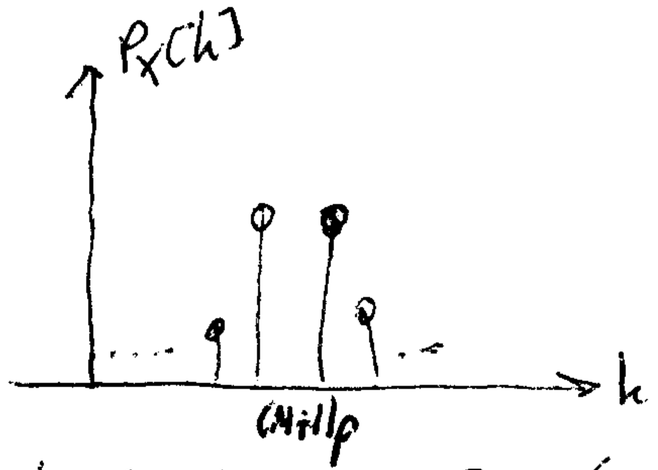
$$= \frac{\frac{M!}{(M-k)!\,k!}}{\frac{M!}{(M-k+1)!\,(k-1)!}} \frac{p^{k-1} \cdot p}{p^{k-1}} \cdot \frac{(1-p)^{M-k}}{(1-p)^{M-k} \cdot (1-p)} = \frac{M-k+1}{k} \frac{p}{1-p}$$

$$= 1 + \frac{(M-k+1)p - k(1-p)}{k(1-p)} = 1 + \frac{(M+1)p - k}{k(1-p)}$$

If $(M+1)p \geq k \Rightarrow Q[k] \geq 1$

If $(M+1)p < k \Rightarrow Q[k] < 1$

$$\frac{P_X((M+1)p)}{P_X((M+1)p - 1)} = 1$$



Maximum occurs at $k = (M+1)p$ and $(M+1)p - 1$. For $(M+1)p$ not an integer, the location of the maximum will be unique or there will be only 1 maximum. ☑

[4] At a party, a large barrel is filled with 99 gag gifts and 1 diamond ring, all enclosed in identical boxes. Each person at the party is given a chance to pick a box from the barrel, open the box to see if the diamond is inside, and if not, close the box and return it to the barrel. What is the probability that at least 19 persons will choose gag gifts before the diamond ring is selected?

Solution:

p = probability of success = 1/100

Geometric law $P_X(k) = (1-p)^{k-1} p = (0.99)^{k-1} \cdot (0.01)$ $k = 1, 2, \ldots$

$$P[\text{at least 19 failures}] = P[k \geq 20] = \sum_{k=20}^{\infty} P_X(k) = \sum_{k=20}^{\infty} (0.99)^{k-1} \cdot (0.01)$$

$$= \frac{(0.01)}{0.99} \sum_{k=20}^{\infty} (0.99)^k = \frac{0.01}{0.99} \frac{(0.99)^{20}}{[1-(0.99)]} = (0.99)^{19} = 0.8262$$

☑

[5]. Prove that the maximum value of a Poisson PMF occurs at $k=[\lambda]$.

Solution: $P_X[k] = \frac{\lambda^k}{k!} e^{-\lambda}$. Let $Q[k] = \frac{P_X(k)}{P_X(k-1)}$

$$Q[k] = \frac{\frac{\lambda^k}{k!} e^{-\lambda}}{\frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda}} = \frac{\lambda^k}{k!} \frac{(k-1)!}{\lambda^{k-1}} = \frac{\lambda^{k-1}\cdot\lambda}{(k-1)!\cdot k} \frac{(k-1)!}{\lambda^{k-1}} = \frac{\lambda}{k}$$

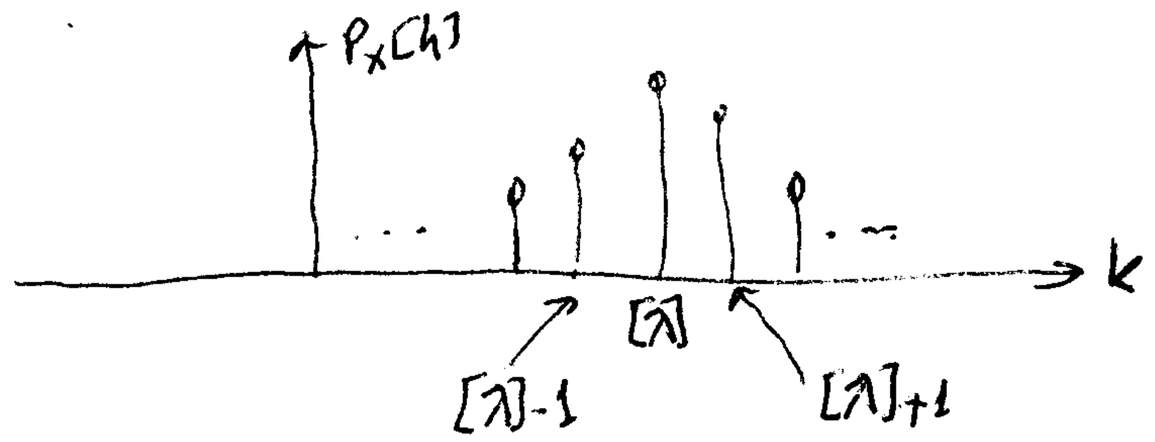
Hence

$$Q[k] \begin{cases} > 1 & \therefore \lambda < k \text{ or } k < \lambda \\ < 1 & \lambda < k \text{ or } k > \lambda \\ = 1 & \lambda = k \end{cases}$$

Maximum occurs at $k=\lambda-1$ and λ if λ is ~~not~~ an integer. Maximum occurs at $k=[\lambda]$ is not an integer because

$$Q[k] = \frac{P_X(k)}{P_X(k-1)} \begin{cases} > 1, & k < \lambda \text{ or } k \le [\lambda] \\ < 1, & k > \lambda \text{ or } k \ge [\lambda]+1 \end{cases}$$

$$\Rightarrow P_X[k] \begin{cases} > P_X[k-1], & k \le [\lambda] \\ < P_X[k-1], & k \ge [\lambda]+1 \end{cases}$$



∎

[6] Prove the following limit: $\lim_{M\to\infty} g(M) = \lim_{M\to\infty} \left(1+\frac{x}{M}\right)^M = e^x$

To do so, not that the same limit is obtained if M is replaced by a continuous variable, say U, and that one can consider $\ln g(u)$ since the logarithm is a continuous function. Hint. Use L' Hospital's rule.

Solution: $f(u) = \left(1+\frac{x}{u}\right)^u \Rightarrow \ln(f(u)) = u \ln\left[1+\frac{x}{u}\right]$

$$\lim_{u\to\infty} \ln(f(u)) = \lim_{u\to\infty} \frac{\ln\left[1+\frac{x}{u}\right]}{1/u}$$

$$= \lim_{u\to\infty} \frac{\frac{1}{1+\frac{x}{u}}(-xu^{-2})}{-u^{-2}} \quad \ldots\ldots \text{(From L' Hospital's rule!)}$$

$$= \lim_{u\to\infty} \frac{x}{1+\frac{x}{u}} = x$$

$\ln f(u) \to x \Rightarrow f(u) \to e^x$ as $u \to \infty$ ∎

[7] A discrete random variable has the PMF

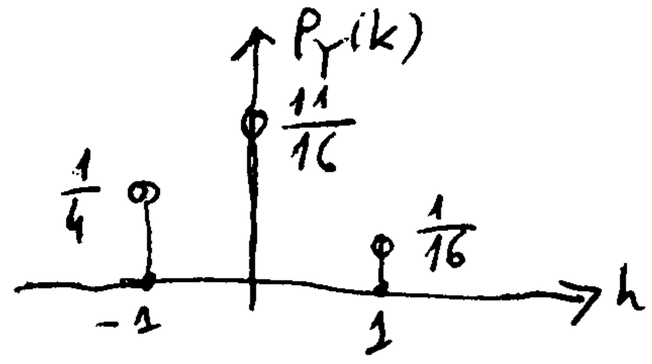
$$P_X(x_i)=\begin{cases}1/2, & x_1=-1\\ 1/4, & x_2=-1/2\\ 1/8, & x_3=0\\ 1/16, & x_4=1/2\\ 1/16, & x_5=1.\end{cases}$$

If $Y=\sin(\pi X)$, find the PMF of Y.

Solution:

X	-1	-1/2	0	1/2	1
Y	0	-1	0	1	0

$$P_Y(0)=P_X[X=-1]+P[X=0]+P_X[X=1]=\frac{1}{2}+\frac{1}{8}+\frac{1}{16}=\frac{11}{16}$$
$$P_Y(-1)=P_X[X=-\tfrac{1}{2}]=\frac{1}{4}$$
$$P_Y(1)=P_X[X=\tfrac{1}{2}]=\frac{1}{16}$$



[8]. Plot the CDF for

$$P_X(x_i)=\begin{cases}1/4, & k=1\\ 1/2, & k=2\\ 1/4, & k=3\end{cases}$$

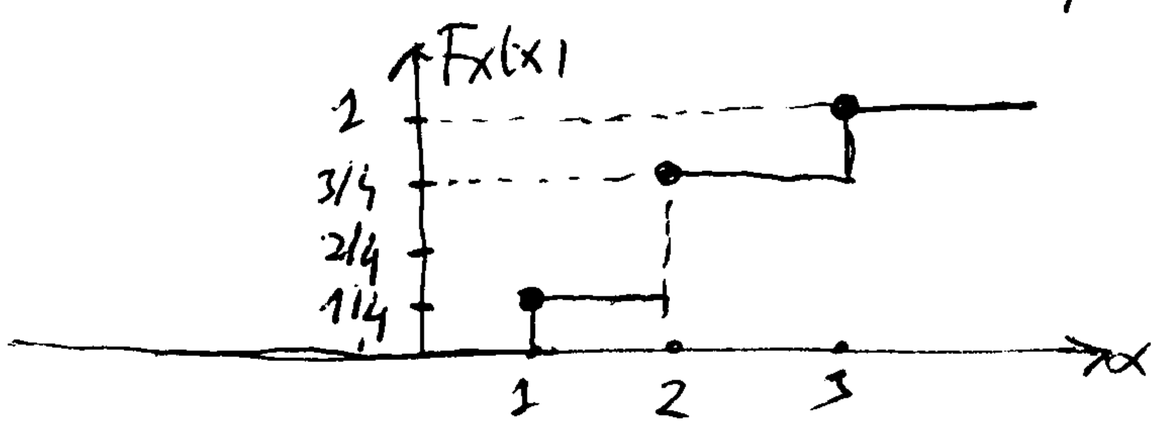
Solution: $F_X(x)=P[X\le x]$

if $-\infty<x<1 \Rightarrow$ $F_X(x)=0$.

if $-\infty\le x<2 \Rightarrow$ $F_X(x)=P[X=1]=1/4$

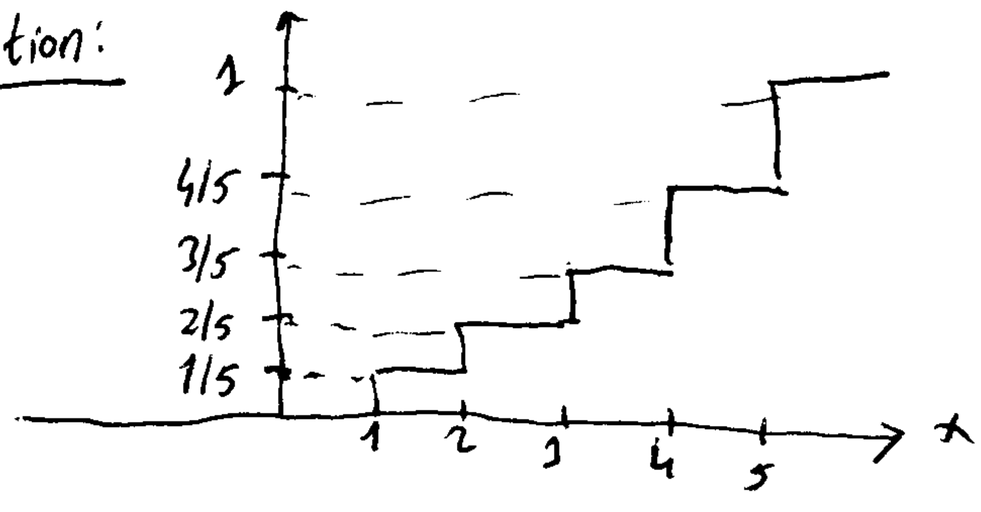
if $-\infty\le x<3 \Rightarrow$ $F_X(x)=P[X=1]+P[X=2]=\frac{1}{4}+\frac{1}{2}=\frac{3}{4}$

if $x\ge 3 \Rightarrow$ $F_X(x)=P[X=1]+P[X=2]+P[X=3]=\frac{1}{4}+\frac{1}{2}+\frac{1}{4}=1$

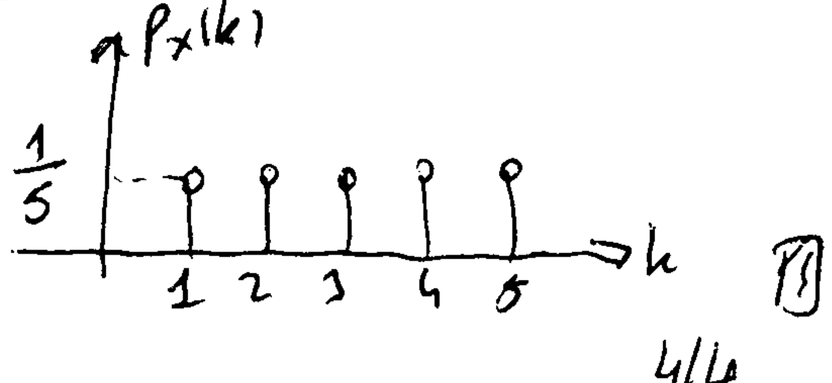


[9] Find the PMF if X is a discrek random variable with the CDF

$$F_X(x)=\begin{cases}0, & x<0\\ [x]/5, & 0\le x\le 5\\ 1, & x>5.\end{cases}$$



$$\boxed{P_X(x)=F_X(x^+)-F_X(x^-)}$$



[8]